



CREST: A causal framework for mitigating shortcut learning in language models through counterfactual reasoning[☆]

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ABSTRACT

Language models excel at numerous natural language processing tasks but often exploit surface patterns rather than develop genuine causal understanding. This limitation leads to vulnerability when encountering out-of-distribution examples or adversarial inputs. We present CREST, a framework incorporating causal learning principles into language models to mitigate shortcut learning through counterfactual reasoning. CREST integrates a causally-informed pre-trained question-answering model with a debiasing framework utilizing counterfactual analysis. The framework explicitly models the causal structure of question-answering tasks and employs controlled interventions to differentiate authentic reasoning pathways from shortcuts. Our multi-branch architecture separates robust causal reasoning from potential shortcut pathways, while the counterfactual reasoning component regulates variable interactions during training and inference. Experiments across six datasets (DREAM, SQuAD, TriviaQA, HotpotQA, TyDi-QA, QuAC) demonstrate CREST achieves 2.41–3.15% improvements over eight baselines, with the strongest gains on multi-hop reasoning. Validation on large language models GPT-J (6B) and Llama-2 (7B) using knowledge editing scenarios shows CREST achieving 44.13% and 45.83% success rates, substantially outperforming Fine-Tuning (19.97–22.97%), ROME (13.67–17.66%), MEMIT (14.05–20.13%), and MeLLO (24.97–29.62%). Critically, CREST demonstrates superior hop-wise accuracy of 30.60% on GPT-J and 60.44% on Llama-2, indicating genuine step-by-step reasoning compared to MeLLO's 0.21–9.90%. CREST exhibits superior adversarial robustness, maintaining 78.4–85.3% performance under attacks compared to 71.2–77.5% for the best baseline. While requiring 30% additional training time, CREST maintains competitive inference speeds with only 3% latency increase, demonstrating practical feasibility for real-world deployment.

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1. Introduction

Pre-trained language models have fundamentally changed natural language processing to capture intricate linguistic patterns from extensive unlabeled text data (Bai et al., 2021; Li et al., 2023; Suzuki et al., 2023). Hence, a language model learns a rich semantic representation of a word or phrase to complete tasks ranging from simple text classification to complex question answering. Recent investigations reveal that good evaluation scores often mask one major drawback: the ability to exploit surface correlations and shortcuts from training data instead of truly learning causal links (Chang et al., 2024; Uribe et al., 2024). Shortcut learning represents a major gap between human-like thinking abilities and present artificial intelligence systems (Dogra et al., 2024; Du et al., 2023; Wang et al., 2025). Humans develop causal conceptual comprehension of concept-linking constructs. On the other hand, language models mostly depend on statistical patterns that work within the training contexts but fail to transfer when confronted with different contexts or adversarial inputs.

Question-answering tasks represent one of the most obvious examples of shortcut learning in language models, as these models can achieve high accuracy while seemingly unable to genuinely identify causal links within the text (Liu et al., 2023; Zhao et al., 2023). For example, models often associate question terms with similar phrases in the text or use option redundancy instead of engaging in the multi-step causal reasoning required for genuine understanding. Such a problem has been addressed by data augmentation or architectural changes in previous works; however, these attempts cure the symptom without addressing the true cause (Cao et al., 2021). The root cause is the failure of the models to distinguish real causal linkages from spurious correlations in their training data. This failure becomes obvious whenever the models face out-of-distribution test cases or adversarial attacks that capitalize on the aforementioned reliance on shortcuts rather than robust causal attributes.

Shortcut learning remains an issue that can be tackled with an appropriate integration of the concepts of causal reasoning into language model system designs and training procedures (Cui et al., 2024; Liu et al., 2024; Wang & Shen, 2024; Zhang et al., 2024). We propose a structural causal model that captures the relationships among the many components of the question-answering task: the passage, question, choices, and the reasoning processes required to get to the correct answer. This causal framework allows one to pick out and intervene on shortcut pathways, forcing the model to depend on proper causal properties rather than spurious correlations. Using counterfactual reasoning, one simulates cases where a certain shortcut is missing, training the model to develop more robust reasoning that corresponds with human-level causal awareness (Ranjbar et al., 2024; Wang et al., 2023). For example, a passage related to renewable energy was given, and the question was, “What percentage of solar energy is mentioned?” A model engaged in shortcut learning would retrieve the first numerical figure close to the term “solar” without any understanding of the actual causal relations between types of energy and their quantitative measuring.” The shortcut negates the need to reason about the relationship ($P \rightarrow R \rightarrow A$) between energy production. It relies on direct lexical matching between passage and answer categories ($P \rightarrow A$), where P describes passage content, R designates reasoning procedures, and A stands for answers.

Going through the paper, the term shortcut learning is retained as the prime candidate for use in describing one of the central problems appearing in various research lines. This problem/subclass has been called surface correlations in research literature, misleading correlations, and spurious correlations. Coming from different vantage points, they all pinpoint the same issue: models learn decision rules that fake something on trivial superficial patterns instead of understanding the causal relation behind it. Surface correlations are statistical relationships designed in the training data with no real causal connection. Misleading correlations are those that appear informative but redirect the learning process away from genuine reasoning. On the other hand, spurious correlations represent false associations correlated with correct answers in training settings, but do not generalize to new contexts (Li et al., 2024). Shortcuts describe any route by which models can seemingly succeed without gaining a robust understanding. These various linguistic approaches manifest the pluriformity of the problem since models could exploit several kinds of superficial patterns depending on the peculiarities of task structure and data features. By conceptualizing these terminologies as different faces of the shortcut learning phenomenon, we will then systematically adduce the solution to the root problem via causal reasoning that discriminates between actual understanding and just pattern exploitation.

The pursuit of robust artificial intelligence systems that can reason about causal relationships represents a fundamental challenge in natural language processing. While current language models achieve impressive performance through statistical pattern recognition, their dependence on superficial correlations limits their ability to generalize beyond training distributions. This limitation becomes particularly problematic in question answering tasks, where genuine understanding requires multi-step causal reasoning rather than pattern matching. Unlike previous modular approaches, our framework establishes formal causal relationships between components through mathematical formulations that enable systematic intervention. The integration operates through principled causal mechanisms rather than empirical stacking, where counterfactual reasoning guides the weighting and interaction of different pathways.

Our paradigm differs from standard methods in improving language model resilience in several fundamental ways. First, unlike techniques that aim to reduce biases through data pretreatment or augmentation, we address the issue deeper by changing how the model learns to reason about relationships in the data. Second, our method preserves model performance while increasing resilience and interpretability, eliminating the traditional trade-off between accuracy and generalization. Third, by including causal reasoning principles, our framework offers a theoretically sound approach to understanding and improving model behavior instead of depending on empirical tricks or heuristics.

The remainder of this paper is organized as follows. Section 2 reviews the related works. Section 3 presents the methodology. Section 4 provides the experiment and results analysis. Finally, Section 5 concludes the paper.

2. Related works

2.1. Language models and question answering

Recent advancements in pre-trained language models have revolutionized question answering capabilities by enhancing their capacity to analyze and comprehend intricate textual linkages (Pouramini & Faili, 2024; Yong et al., 2023; Zeng et al., 2023). In Cao and Liu (2023), authors proposed a novel model ReLMKG: reasoning with pre-trained language models and knowledge graphs for complex question answering. In He et al. (2025), authors presented PTM4Tag+, a tag recommendation framework for Stack Overflow posts that utilize PTMs in language modeling. In Yang et al. (2024), authors proposed multiscale attention pre-training model (MAPM) to enhance multimodal feature fusion. In Lim et al. (2024), authors proposed a novel three-stage framework named UniRaG for tackling multimodal question answering, which encompassed unified knowledge representation, context retrieval, and answer generation. In Li et al. (2022), authors presented a pre-trainable universal encoder-decoder network (Uni-EDEN) to facilitate both vision-language perception (e.g., visual question answering) and generation (e.g., image captioning). In Pouramini and Faili (2024), authors introduced the match task to objective (MTO) framework and methods for determining the appropriate objective for a given task. In Kang et al. (2024), authors proposed an inhibition adaptation (InA) fine-tuning method that aims to reduce the number of added tunable weights and appropriately reweight knowledge derived from pre-trained language models. In Duan et al. (2024), authors proposed a lightweight expression-enhanced large-scale pre-trained language model, called EBERT, for mathematics education. These models demonstrate remarkable performance across various question-answering formats, from span-based extraction to multiple-choice reasoning (Choi et al., 2022; Peng et al., 2024). However, deeper analysis reveals that strong performance often masks fundamental limitations in causal understanding, where models exploit surface patterns rather than developing genuine reasoning capabilities.

The unique causal shortcomings in question-answering tasks manifest through several pathways: temporal reasoning failures, where models cannot track causal sequences across sentences; counterfactual blindness, where models struggle with hypothetical scenarios; and attribution errors, where models incorrectly identify cause-effect relationships between entities. These limitations result in brittleness when encountering questions that require genuine causal inference rather than pattern recognition.

2.2. Shortcut learning in neural networks

Shortcut learning is a critical challenge in deep learning systems, wherein models utilize simplistic decision rules that perform adequately on conventional test sets but do not grasp the essential linkages required for effective generalization (Chuah et al., 2023; Sui et al., 2024). This difficulty is especially pronounced in language comprehension tasks, where the high dimensionality of text offers several chances for models to acquire misleading associations. Studies indicate that models frequently circumvent appropriate thinking processes in favor of superficial patterns (Du et al., 2023). In question answering, models may learn to associate question terms with like phrases in the passage without comprehending the underlying semantic linkages. These shortcuts result in fragile performance when models confront out-of-distribution instances or endure hostile assaults.

2.3. Causal learning in natural language processing

Causal learning has arisen as a viable approach for enhancing the resilience and reliability of natural language processing systems (Busto-Castiñeira et al., 2024; Feder et al., 2022; Tiwari & Zhang, 2024). Causal techniques represent the links between various components of language processing tasks, enabling models to differentiate between authentic semantic connections and misleading correlations. Recent research has investigated many methods for integrating causal reasoning into language models, encompassing counterfactual data augmentation (Ferrario & Loi, 2022), causal intervention during training (Morsanyi et al., 2024), and deliberately organized causal structures (Sadeghi & Soo, 2022). These strategies demonstrate the potential to assist models in cultivating more resilient characteristics that accurately represent genuine causal links in language comprehension challenges.

Recent developments have introduced automated methods for counterfactual data augmentation, including large language model-based generation and predefined counterfactual dictionaries. Additionally, active learning-based causal debiasing methods have emerged to identify causal biases without manual specification automatically. However, these approaches often lack the systematic causal intervention capabilities required for robust shortcut mitigation in complex reasoning tasks. LLM-based augmentation may introduce new biases through the generation process, while dictionary-based methods remain limited to predefined patterns (Si et al., 2024; Wu et al., 2024). Active learning approaches, although promising for bias discovery, typically focus on identifying rather than systematically intervening in causal relationships (Du et al., 2024).

The current methodologies for enhancing question answering systems exhibit several constraints. Data augmentation techniques need considerable manual labor and may lack effective generalization across various shortcut kinds. Architectural alterations frequently target certain prejudices without offering a holistic remedy. Conventional debiasing methods concentrate on eliminating recognized biases but may fail to address the complete spectrum of spurious correlations that models might acquire. Conversely, our proposed system mitigates these constraints by integrating causal reasoning concepts directly into the model design and training process. Our method facilitates the systematic identification and mitigation of shortcut learning by explicitly modeling and interfering in causal links while maintaining valid reasoning pathways. This causal viewpoint offers a theoretically sound approach to enhance model robustness and interpretability, resulting in more dependable question-answering systems that reflect genuine language comprehension abilities more accurately.

3. Method

This section outlines our methodology for incorporating causal reasoning into pre-trained language models to mitigate shortcut learning. Our methodology integrates causal intervention with counterfactual reasoning to improve model robustness and interpretability.

To illustrate our methodology, consider a reading comprehension model that learns to associate question words with passage locations rather than understanding semantic content—a classic shortcut. Our framework addresses this by implementing three interconnected steps. First, we construct a causal graph where passage semantics mediates the relationship between questions and answers, explicitly modeling how genuine understanding should flow from question interpretation through semantic analysis to answer selection. Second, we implement counterfactual reasoning that tests whether the model's predictions change appropriately when the semantic content varies. At the same time, surface patterns remain constant—for instance, when we change the semantic meaning of a passage while keeping similar vocabulary. Third, we use a multi-branch architecture to explicitly separate shortcut-prone features from causal reasoning pathways, allowing the model to develop a robust understanding while monitoring potential vulnerabilities. This three-step process ensures the model develops authentic comprehension rather than superficial pattern matching.

3.1. Problem definition

In the context of language model learning, causal relationships can be formalized through a directed acyclic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where vertices $\mathcal{V}$ represent system variables and edges $\mathcal{E}$ encode their causal dependencies. Within this framework, we define causal intervention as a formal operation that breaks the causal dependencies flowing into a variable, allowing us to observe the effect of setting that variable to a specific value while holding the causal structure constant, particularly in identifying and mitigating shortcut learning behaviors in transformer-based language models.

When considering causal interventions in language understanding tasks, let $\mathcal{X}$ represent input features (such as text embeddings or attention patterns), $\mathcal{Y}$ denote target outputs, and $\mathcal{M}$ indicate mediating variables that could enable shortcut learning (Amini et al., 2023). A fundamental causal intervention can be expressed through the *do*-operator:

$$P(\mathcal{Y} \mid do(\mathcal{X})) = P(\mathcal{Y} \mid \mathcal{X} = \tilde{x}) = f(\tilde{x}) \quad (1)$$

where $\tilde{x}$ represents intervened values and $f(\cdot)$ encapsulates the transformed relationship. The upper part of Fig. 1 illustrates this intervention framework in the context of language model learning:

For question answering with language models, we develop a structural causal model incorporating input passage I , query Q , candidate responses C , reasoning processes H , and final answer A . The causal relationships between these variables can be expressed through:

$$\mathcal{A}_{I,Q,C,H} = \mathcal{A}(I = i, Q = q, C = c, H = h) \quad (2)$$

When we introduce controlled interventions where all variables (including hidden states) are set to null (denoted as $*$), we obtain our base intervention case:

$$\mathcal{A}_{I^*,Q^*,C^*,H^*} = \mathcal{A}(I = i^*, Q = q^*, C = c^*, H = h^*) \quad (3)$$

While the lower part of Fig. 1 presents the complete structural causal model for language model-based question answering. The causal graph construction follows domain knowledge about question-answering tasks, where passages provide evidence, questions specify information needs, and answers represent target outputs. The mediating variables capture intermediate reasoning states that can enable shortcuts. This construction process relies on understanding the intended causal flow in reading comprehension tasks.

The input passage indeed acts as a potential confounder, creating spurious correlations between questions and answers through shared lexical content. Our framework addresses this by explicitly modeling these confounding relationships and using counterfactual interventions to break spurious pathways while preserving genuine causal connections. The multi-branch architecture separates these pathways for independent analysis.

Consider an example where the passage discusses renewable energy policies; the question asks, “What percentage of energy comes from solar power?” and options include (A) 15%, (B) 23%, (C) 8%, (D) Cannot be determined. In our causal model, I represents the passage embeddings containing policy information, Q encodes the percentage inquiry, C contains the numerical options, H captures the reasoning process that must locate and interpret the relevant statistic, and A represents the final answer selection. A shortcut pathway might directly match “solar” and “percentage” terms without understanding the causal relationship between policy implementation and energy statistics. Our intervention $do(Q = \emptyset)$ removes question information to isolate whether the model relies on genuine passage comprehension versus simple keyword matching, revealing spurious correlations that bypass authentic reasoning processes.

Then, for counterfactual reasoning, we create scenarios where some variables are intervened in while others maintain their original values. Specifically, we keep the hidden state unmodified while intervening on input variables:

$$\mathcal{A}_{\text{input}^*} = \mathcal{A}(I = \emptyset, Q = \emptyset, C = \emptyset, H = h) \quad (4)$$

This enables us to analyze the natural indirect effect (NIE):

$$\text{NIE} = \mathcal{A}(I'Q'C'h) - \mathcal{A}(I'Q'C'h^*) \quad (5)$$

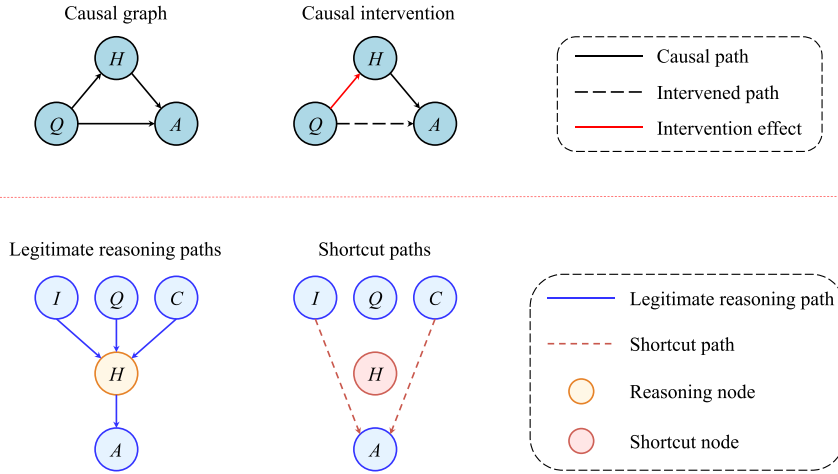


Fig. 1. Causal framework for question answering.

where I' denotes intervened input passage representations, Q' indicates modified query encodings, C' represents altered choice option embeddings, h captures preserved hidden states from original processing, h^* denotes nullified hidden states under intervention, and $\mathcal{A}(\cdot)$ represents the answer prediction function. The NIE quantifies how mediating variables influence final predictions when input variables are controlled through causal intervention.

The controlled indirect effect (CIE):

$$\text{CIE} = \mathcal{A}(I, Q, C, H) - \mathcal{A}(I, Q, C, H^*) \quad (6)$$

Using this formal causal framework, shortcut learning in language models may be identified systematically and prevented by examining how the various components interact and influence the final predictions. This distinction leads to identifying real language understanding and mere pattern matching and building more robust and reliable question answering procedures.

3.2. Model overview

In our methodology, we address shortcut learning in language models by using causal reasoning and bias reduction techniques. The architecture is split into two branches: the first branch is the causally-informed pre-trained question-answering model, and the second branch is the debiasing framework that works counterfactually.

The first component uses causal inference to distinguish between genuine reasoning pathways and shortcut learning tendencies in transformer-based language models. By carefully designing the model architecture and training methodology, our method enables models to pay attention to true causal relationships, while discouraging attention to spurious correlations, which are quite frequently considered by conventional approaches. The separation happens through a multi-branch architecture, where each branch scrutinizes the input data from different causal perspectives. Consider the multi-branch architecture as several parallel investigation teams looking at the same evidence from different analytical perspectives. Information entering the system flows simultaneously through two paths that instantiate conceptually distinct reasoning styles. The robust causal branch reasons over the incoming information by building a structured representation of cause-and-effect relationships. This pathway maps concepts onto meaningful connections to check if the question requirements are semantically related to the contents of the passage and if the answer choices fairly implement the intended information need. The causal view precedes white-box semantic and logical coherence rather than statistical correlations. At the same time, the shortcut branch works on the same incoming information via pattern-recognition mechanisms that identify superficial correlations likely to be exploited by models. Shortcut detection consists of lexical overlap between question and answer, keyword matches on the surface, or statistical regularities that can give correct answers without genuine understanding. Thus, this explicit bifurcation allows the system to combine the development of actual causal reasoning with actively monitoring for shortcuts. Instead of attempting to suppress these unwanted behaviors, the design offers CREST the potential to explicitly represent and learn both pathways while developing a preference for robust causal reasoning. This multi-branch design also offers transparency into the model's decisions by showing whether predictions came from real reasoning or pattern exploitation. Fig. 2 depicts the granular steps of the causal reasoning flow.

In Fig. 2, the causal feature extractor examines semantic information with an attentional mechanism, detecting meaningful relations between concepts rather than mere word matches on the surface level. It translates the raw input text embeddings into representation spaces that express causal dependencies: the concept of “policy changes” causing the appearance of “energy statistics” instead of merely co-occurring. The counterfactual generator masks or alters input fragments to generate alternate scenarios, asking, “What would happen if this piece of information were absent?” This counterfactual analysis helps in discerning real model dependencies from mere surface correlations. After combining the output of the causal analysis with learned attention weights, the

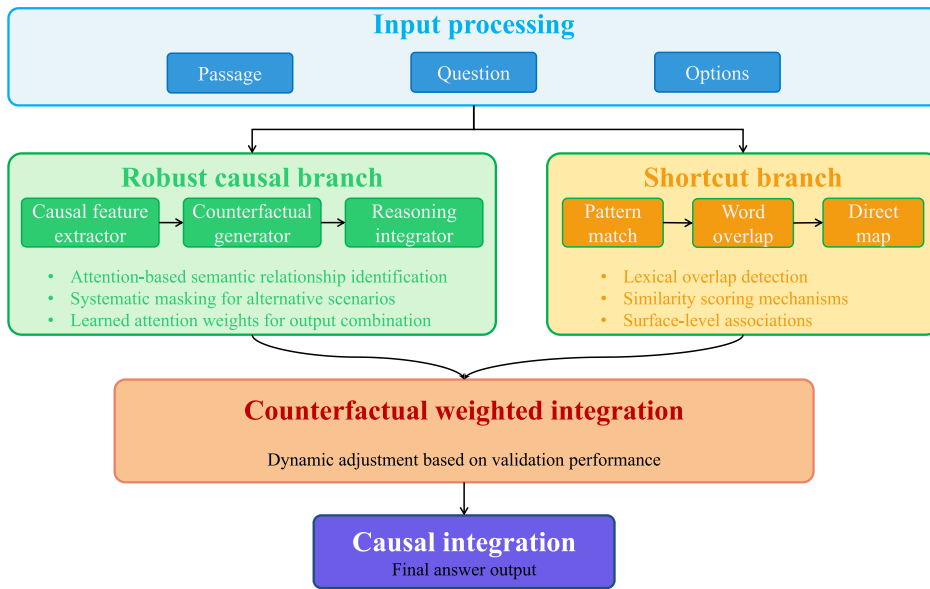


Fig. 2. Framework architecture.

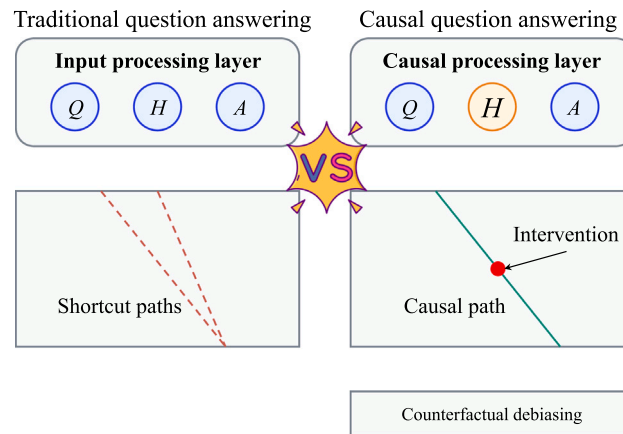


Fig. 3. Comparison between traditional and causal frameworks.

reasoning integration uses a weighting system that favors real causal evidence over shortcuts. Meanwhile, the shortcut branch goes through Pattern match (detecting lexical overlaps), word overlap (something like similarity scores), and Direct map (surface-level association between questions and passages). The counterfactual weighted integration manages custom trade-offs between these paths, on validation learning to prioritize causal reasoning yet keep shortcut detection standing by, thus forming a system that can locate but resist spurious patterns.

The causal-aware pre-trained model implements counterfactual reasoning in conditioning the interaction between variables throughout training and inference. When given input text, the model first generates causal representations explaining the relationships among different textual components within the text. These representations are then carried through parallel processing branches, one being rigorously causally reasoned over and the others simulating potential shortcut pathways. By explicitly contrasting these different pathways, we can currently analyze and control the predictive mechanism of the model.

Our debiasing framework is predicated on this causal foundation and incorporates targeted interventions to mitigate prevalent biases in question-answering systems. Instead of perceiving bias as a singular issue, we analyze it by breaking it down into distinct patterns that can be effectively addressed through counterfactual reasoning. This methodology enables us to uphold the model's performance on valid reasoning pathways while methodically eliminating undesirable shortcuts. Fig. 3 illustrates the comparison between traditional and causal frameworks.

In Fig. 3, 'H' represents reasoning processes that mediate between questions and answers. The traditional framework shows 'H' as a simple, intermediate step that often relies on pattern matching, while the causal framework depicts 'R' as structured

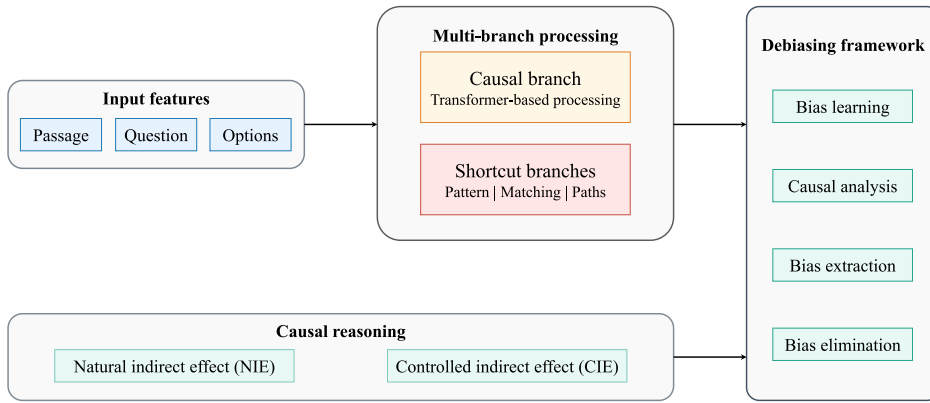


Fig. 4. Complete model architecture with causal learning and debiasing components.

reasoning that maintains causal consistency. The two dotted lines in the traditional system represent potential shortcut pathways that bypass genuine reasoning by directly connecting questions to answers through surface patterns. These shortcuts allow models to achieve the correct answers without developing a true understanding. The causal framework eliminates these shortcuts through intervention mechanisms, forcing all processing through the structured reasoning component ‘H,’ which ensures that answers derive from authentic causal analysis rather than spurious correlations between question terms and passage content.

An essential advancement in our methodology is the management of the interplay between causal reasoning pathways and possible shortcuts. Conventional approaches frequently encounter difficulties differentiating between authentic patterns and misleading correlations, resulting in models that may excel on established benchmarks but lack generalizability. Our methodology mitigates this constraint by explicitly modeling the causal structure of the question-answering task, enabling us to detect and act on shorter pathways while maintaining valid reasoning chains. Fig. 4 depicts the complete model architecture with causal learning and debiasing components.

Fig. 4 shows the integration of causal learning and bias elimination components into a unified framework. The multi-branch processing separates causal reasoning from shortcut detection, so the system develops a strong understanding while monitoring vulnerabilities. The causal analysis identifies the authentic relations among concepts, while the bias extraction mechanisms find and isolate spurious patterns that models might exploit. The bias removal actively neutralizes those spurious correlations on the grounds of counterfactual reasoning. The mathematical definitions of NIE and CIE quantify pathway-level effects on the end predictions, which the system employs to control the degree of true reasoning against shortcut exploitation activities throughout learning.

The training involves multiple meticulously structured phases that gradually impart causal comprehension to the model while reducing reliance on heuristics. In the first stage of training, it is conceivable that the model learns both causal and shortcut attributes; however, we apply counterfactual regularization to instill a preference for robust causal patterns within the machine. This is realized via a novel loss function that balances performance on original instances versus behavior on counterfactually treated inputs.

Our paradigm trains the existing methodological apparatus in several relevant technical aspects. First, contrary to those methodologies working around shortcuts via data augmentations or architectural constraints, in our case, the problem shall be tackled from the beginning by modeling the causal structure of the job. Secondly, our method improves without sacrificing the model’s accuracy, strengthening the trade-off for generalized interpretations. The more generalized an algorithm becomes, its prediction accuracy suffers, and vice versa. Third, based on a causal perspective, our framework provides a more formal and sound venue for analyzing and improving the performance of a given method than mere experimental analysis or heuristic considerations.

Debiasing is integral to the main model, embedding counterfactual scenarios during training and inference. Hence, the model can spot and fix biased predictions without requiring expensive retraining or altering its architecture. This system makes adaptability to new biases a huge plus, leaving behind traditional approaches that cater to only certain bias patterns.

3.3. Causal pre-trained question answering model

Our causal pre-trained question answering model presents a method integrating a multi-branch architecture with counterfactual reasoning to mitigate shortcut learning in language models. The framework is based on transformer-based architectures and integrates explicit causal reasoning routes.

In our multi-branch training system, let $\mathcal{T}$ denote the transformer encoder that processes input sequences. The input space comprises context passages $\mathcal{D}$, queries $\mathcal{Q}$, and option sets $\mathcal{O}$. The transformer encoding procedure may be articulated as follows:

$$\mathcal{H} = \mathcal{T} ([\mathcal{D}; \mathcal{Q}; \mathcal{O}]; \theta_{\mathcal{T}}) \quad (7)$$

where $[\cdot; \cdot; \cdot]$ denotes concatenation and $\theta_{\mathcal{T}}$ represents the transformer parameters. The hidden representations $\mathcal{H}$ then flow through parallel processing branches.

The robust causal branch processes complete inputs through a causal reasoning module C :

$$\mathcal{Z}_c = C(\mathcal{H}; \theta_c) \quad (8)$$

For shortcut branches $n \in 1, \dots, N$, we process subsets of inputs S_n :

$$\mathcal{Z}_n^s = S_n(\mathcal{H}; \theta_n^s) \quad (9)$$

The integration of these branches occurs through a causal ensemble mechanism:

$$P(\mathcal{A} | \mathcal{X}) = \lambda \mathcal{P}_c(\mathcal{A} | \mathcal{Z}_c) + \sum_{n=1}^N \mu_n \mathcal{P}_n^s(\mathcal{A} | \mathcal{Z}_n^s) \quad (10)$$

where λ and μ_n represent learned weights subject to normalization constraints:

$$\lambda + \sum_{n=1}^N \mu_n = 1, \lambda, \mu_n \geq 0 \quad (11)$$

During training, we employ a counterfactual-aware loss function:

$$\mathcal{L} = \mathcal{L}_{CE}(\mathcal{P}(\mathcal{A} | \mathcal{X}), y) + \gamma \sum_{n=1}^N \mathcal{L}_{CF}(\mathcal{P}_n^s(\mathcal{A} | \mathcal{Z}_n^s), \mathcal{P}_c(\mathcal{A} | \mathcal{Z}_c)) \quad (12)$$

where $\mathcal{L}_{CE}$ represents the standard cross-entropy loss, and $\mathcal{L}_{CF}$ denotes the counterfactual consistency loss defined as:

$$\mathcal{L}_{CF}(\mathcal{P}_n^s, \mathcal{P}_c) = \|\mathcal{P}_n^s - \mathcal{P}_c\|_2 + \alpha \cdot \text{KL}(\mathcal{P}_n^s \| \mathcal{P}_c) \quad (13)$$

This formulation measures the divergence between shortcut predictions $\mathcal{P}_n^s$ and causal predictions $\mathcal{P}_c$. The L2 norm captures direct prediction differences, whereas the Kullback–Leibler divergence quantifies distributional discrepancies between the two reasoning pathways. The parameter γ controls the relative importance of counterfactual regularization within the overall training objective, balancing standard task performance against causal consistency requirements.

This counterfactual-aware loss function serves as the foundation for our training objective. We expand this formulation during implementation to include additional regularization terms for practical deployment. The complete training objective becomes

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CE} + \gamma \sum \mathcal{L}_{CF} + \lambda_1 \mathcal{L}_{\text{causal}} + \lambda_2 \mathcal{L}_{\text{counter}} + \lambda_3 \mathcal{L}_{\text{reg}} \quad (14)$$

where $\mathcal{L}_{CE}$ represents the base cross-entropy loss, $\gamma \sum \mathcal{L}_{CF}$ corresponds to our counterfactual consistency term from Eq. (12), and the additional terms provide stability during training. The causal consistency term $\mathcal{L}_{\text{causal}} = \text{KL}(\mathcal{P}_{\text{causal}} \| \mathcal{P}_{\text{shortcut}}) + \|\mathcal{Z}_c - \mathcal{Z}_n^s\|_2$ measures the alignment between the representations of the causal and shortcut branches. The counterfactual stability term $\mathcal{L}_{\text{counter}} = \|P(\mathcal{A}|\text{original}) - P(\mathcal{A}|\text{counterfactual})\|_2$ ensures consistent predictions under intervention. The regularization term $\mathcal{L}_{\text{reg}} = \sum_i \theta_i^2$ applies the standard L2 weight decay. The weights $\lambda_1, \lambda_2, \lambda_3$ are empirically determined by validation.

The counterfactual stability term $\mathcal{L}_{\text{counter}}$ directly supports causal reasoning objective by encouraging the model to maintain consistent decision-making processes when exposed to interventions that should not affect the outcome. This connects to NIE and CIE analysis by ensuring that the model learns to distinguish between legitimate causal pathways and spurious correlations. When we intervene on irrelevant variables, a model relying on genuine causal understanding should produce similar predictions, while a model dependent on shortcuts will show unstable behavior. By minimizing prediction variance under such interventions, $\mathcal{L}_{\text{counter}}$ reinforces the causal pathways identified through NIE and CIE decomposition.

Fig. 5 demonstrates intervention effects using validation samples from the DREAM dataset. The analysis employs 1000 randomly selected question–answer pairs, where interventions are applied systematically across different variable combinations. The probability distributions show the magnitude of causal effects under controlled experimental conditions.

The counterfactual reasoning component is crucial in identifying and mitigating shortcut learning. The counterfactual reasoning component is crucial in identifying and mitigating shortcut learning. We define two control mechanisms that we collectively refer to as counterfactual variable control (CVC): input variable counterfactual control (IVC) and mediator variable counterfactual control (MVC). For IVC, we intervene on input variables while maintaining mediator states:

$$\mathcal{A}(\mathcal{D} \cdot \mathcal{Q} \cdot \mathcal{O}^*, h) = \mathcal{A}(\emptyset_{\mathcal{D}}, \emptyset_{\mathcal{Q}}, \emptyset_{\mathcal{O}}, h) \quad (15)$$

For MVC, we maintain input values while intervening on mediator states:

$$\mathcal{A}(\mathcal{D}, \mathcal{Q}, \mathcal{O}, h^*) = \mathcal{A}(d, q, o, \emptyset_h) \quad (16)$$

Our intervention strategies follow Pearl’s causal hierarchy, where IVC and MVC implement do-calculus operations. The masking operations represent controlled interventions that break causal pathways systematically. For IVC, setting input variables to null ($\emptyset$) simulates scenarios where specific input channels are blocked, enabling measurement of mediator-dependent effects. For MVC, maintaining inputs while nullifying mediators isolate direct causal pathways from confounded routes.

The IVC approach isolates mediator effects by examining how hidden states influence outcomes when inputs are controlled. While inputs serve as direct causes, their intervention reveals bias pathways that operate through mediating variables. The MVC approach examines direct input–output relationships by controlling mediator states. These interventions, while not perfectly mimicking natural co-occurrence, provide bounded estimates of causal effects under controlled experimental conditions.

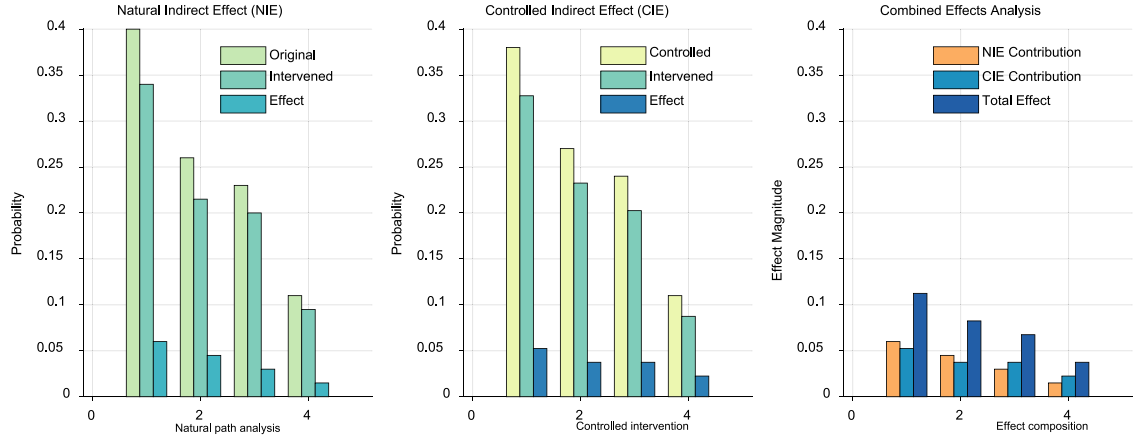


Fig. 5. Counterfactual reasoning and intervention effects.

Table 1

Key variables and their roles in causal inference.

Variable	Description	Role in inference
$\mathcal{T}$	Transformer encoder	Primary text processing component
$\mathcal{D}$	Context passage	Input document for reasoning
$\mathcal{Q}$	Query text	Question or prompt for analysis
$\mathcal{O}$	Option set	Candidate answer choices
$\mathcal{H}$	Reasoning processes	Intermediate representations
$\theta_{\mathcal{T}}$	Model parameters	Transformer configuration values
$\mathcal{Z}_c$	Causal features	Robust reasoning pathways
$\mathcal{Z}_n^s$	Shortcut features	Pattern-matching shortcuts
λ	Causal weight	Importance of causal reasoning
μ_n	Shortcut weights	Impact of shortcut paths
$\mathcal{A}(\cdot)$	Answer function	Prediction mechanism for QA task
$\emptyset_D$	Nullified passage	Intervened document state
$\emptyset_Q$	Nullified query	Intervened question state
$\emptyset_O$	Nullified options	Intervened choice state
h^*	Intervened hidden states	Controlled mediator variables
d, q, o	Original inputs	Unmodified input representations
$\emptyset_h$	Nullified mediators	Controlled hidden state intervention
$\mathcal{A}_{IVC}$	IVC prediction	Input variable counterfactual output
$\mathcal{A}_{MVC}$	MVC prediction	Mediator variable counterfactual output
$\mathcal{A}_{final}$	Final prediction	Combined counterfactual inference
α	IVC weight	Scaling parameter for input intervention
β	MVC weight	Scaling parameter for mediator intervention
η_1	Optimization parameter	Grid search variable for validation
η_2	Optimization parameter	Grid search variable for validation
$\text{Perf}(\cdot)$	Performance function	Evaluation metric on validation set
$\mathcal{D}_{val}$	Validation dataset	Data for parameter optimization
$\hat{\mathcal{A}}$	Bias-adjusted prediction	Debiased model output

Table 1 shows the key variables and their meanings in inference process.

Through these interventions, we can compute the NIE:

$$\text{NIE} = \mathcal{A}(D'Q'O'h) - \mathcal{A}(D'Q'O'\emptyset_h) \quad (17)$$

and the CIE:

$$\text{CIE} = \mathcal{A}(d, q, o, h) - \mathcal{A}(d, q, o, \emptyset_h) \quad (18)$$

The counterfactual predictions provide insights into model behavior under different causal scenarios. For prediction, we combine information from both IVC and MVC:

$$\mathcal{A}_{final} = \alpha \mathcal{A}_{IVC} + \beta \mathcal{A}_{MVC} \quad (19)$$

where α and β are learned through validation:

$$(\eta_1, \eta_2) = \arg \max_{\eta_1, \eta_2} \text{Perf}(\mathcal{D}_{val}, \hat{\mathcal{A}}(\eta_1, \eta_2)) \quad (20)$$

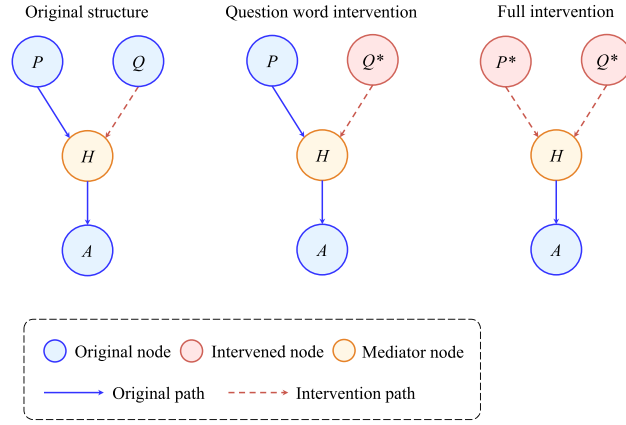


Fig. 6. Causal graph construction.

This framework facilitates explicitly modeling causal linkages and identifying and mitigating shortcut learning tendencies in language models. The multi-branch design and counterfactual reasoning offer a theoretically sound method for enhancing model resilience and interpretability.

3.4. Causal debiasing framework

Our causal debiasing framework presents a systematic method for mitigating intrinsic biases in language model reasoning. In contrast to conventional debiasing techniques that emphasize data alteration or modifications to model design, we utilize causal theory to detect and alleviate biases using counterfactual reasoning.

The methodology starts with bias learning, enabling the foundational language model $\mathcal{M}$ to acquire possible biases during its preliminary training. This transpires via normal cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{n} \sum_{i=1}^n \sum_{y \in \mathcal{Y}} \pi_{i,y} \log(\hat{\pi}_{i,y}) \quad (21)$$

where ϕ represents model parameters, π_i denotes true label distributions, and $\hat{\pi}_i$ indicates predicted probabilities. The bias learning phase allows us to understand how the model naturally develops shortcuts and biases.

To analyze and intervene on these biases, we construct a causal graph that captures key relationships between linguistic elements, as shown in Fig. 6.

Fig. 6 depicts nodes that symbolize textual elements: D (documents), Q (queries), $\mathcal{O}$ (options), with the mediating variable H (hidden states) and the target variable $\mathcal{A}$ (answers). We recognize two principal categories of bias: query-term alignment and option redundancy.

For query-term matching bias, we construct counterfactual scenarios by masking non-question terms:

$$Q^* = \text{MASK}(Q \setminus Q_w) \quad (22)$$

where Q_w represents question words and MASK removes other terms. The corresponding counterfactual prediction becomes:

$$\mathcal{A}(Q^*) = f(D, \emptyset_Q, \mathcal{O}) = P(\mathcal{A} \mid \text{do}(Q = \emptyset_Q)) \quad (23)$$

For option overlap bias, we construct counterfactuals by masking both document and query:

$$(D^*, Q^*) = \text{MASK}(D, Q) \quad (24)$$

The bias-adjusted prediction integrates counterfactual outcomes through:

$$\hat{\mathcal{A}} = \mathcal{A} - \eta_1 \mathcal{A}(Q^*) - \eta_2 \mathcal{A}(D^*, Q^*) \quad (25)$$

where η_1 and η_2 are learned scaling factors that determine the influence of each bias type. These factors are optimized through a grid search on a validation set:

$$(\eta_1^*, \eta_2^*) = \arg \max_{\eta_1, \eta_2} \text{Perf}(D_{\text{val}}, \hat{\mathcal{A}}(\eta_1, \eta_2)) \quad (26)$$

Fig. 7 demonstrates the parameter optimization process using validation samples from the DREAM dataset with 1000 randomly selected question-answer pairs. The left panel shows a parameter optimization grid where η_1 and η_2 represent scaling factors that control question-word bias and option-overlap bias, respectively. The contour plot reveals the optimization landscape, with the red

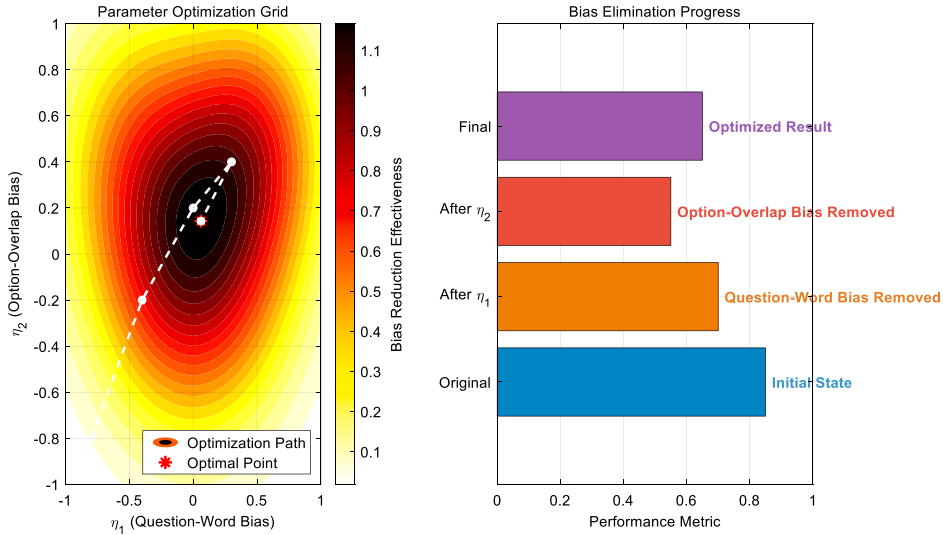


Fig. 7. Parameter optimization grid for bias elimination.

path indicating the search trajectory toward optimal values. The right panel displays bias elimination progress, showing performance improvements as different bias types are systematically removed. The optimization constrains $\eta_1, \eta_2 \in [-1, 1]$ to ensure stability, with the optimal point representing the configuration that maximizes validation performance while minimizing bias dependence. The visualization demonstrates how grid search systematically explores the parameter space to find the optimal balance between bias mitigation and task performance.

The optimization process ensures stability by constraining the scaling factors $\eta_1, \eta_2 \in [-1, 1]$. To identify bias patterns systematically, we employ a bias scoring mechanism:

$$\text{BiasScore}(x) = \mathbb{E}_{c \sim C} [D_{\text{KL}}(P(\mathcal{A} | x) \| P(\mathcal{A} | cf))] \quad (27)$$

where D_{KL} represents the Kullback–Leibler divergence and C denotes the set of counterfactual examples for input x . This score helps quantify how much the model relies on spurious correlations rather than causal reasoning.

The debiasing process involves dynamic weight adjustment during inference:

$$\omega_t = \omega_{t-1} \odot \exp(-\gamma \cdot \text{BiasScore}(x_t)) \quad (28)$$

where ω_t represents bias-adjustment weights at step t , and γ controls the adaptation rate. The final prediction aggregates information across multiple counterfactual scenarios:

$$\mathcal{A}_{\text{final}} = \text{softmax}(\lambda \mathcal{A}_{\text{causal}} + (1 - \lambda) \mathcal{A}_{\text{debiased}}) \quad (29)$$

where $\mathcal{A}_{\text{causal}}$ represents predictions from the robust causal branch after counterfactual processing, and $\mathcal{A}_{\text{debiased}}$ denotes outputs from the bias mitigation framework.

To obtain $\mathcal{A}_{\text{causal}}$: Forward pass input through the robust causal branch, apply the causal feature extractor $C(H; \theta_c)$, generate counterfactual scenarios via the counterfactual generator, then aggregate outputs through the reasoning integrator using learned attention weights. The result represents predictions derived from authentic causal relationships. To obtain $\mathcal{A}_{\text{debiased}}$: Process input through the bias mitigation framework by constructing counterfactual scenarios using $\mathcal{A}(Q) = f(P, \theta, O)$ and $\mathcal{A}(D, Q) = \text{MASK}(D, Q)$, then compute bias-adjusted predictions via $\hat{A} = A - \eta_1 \mathcal{A}(Q) - \eta_2 \mathcal{A}(D, Q)$ where η_1, η_2 are optimized scaling factors.

The effectiveness of our framework arises from its ability to de-bias data through causal intervention, addressing both explicit and implicit biases. Counterfactual situations provide an understanding of the model's behavior when compelled to rely on different features, thus leading to targeted debiasing of these features while ensuring that the methods employed do not damage the overall model accuracy.

This method can exploit the powers of classical methods, offering a few important benefits in comparison with traditional ones. First, it is an inference-level framework and does not require the base model to have any alterations in its architecture or training procedure. Second, it is adaptable to other biases as and when they arise in the system by formulating apt counterfactual scenarios. Third, and most crucially, it promotes interpretability by presenting a transparent view of factors on which biases influence model predictions.

Utilizing this causal framework, we can systematically identify and address the diverse manifestations of shortcut learning within language models. The integration of counterfactual reasoning with dynamic bias adjustment facilitates the development of more resilient and equitable question-answering systems while maintaining authentic reasoning abilities.

4. Experiment

4.1. Experimental setup

Experiments were performed to test the applicability of our causal learning framework across various dimensions. The evaluation used six well-known datasets: DREAM, SQuAD, TriviaQA, HotpotQA, TyDi-QA, and QuAC, each of which provides different attributes on which to judge various dimensions of language model performance (Mao et al., 2024; Sinthusha et al., 2025). DREAM consists of 10,197 multiple-choice questions from 6,444 dialogs, designed for dialog-based reading comprehension. SQuAD is span-based question answering based on Wikipedia articles, while TriviaQA consists of more than 650,000 triples of question-answer-evidence, along with evidence documents sourced independently. HotpotQA is a question-answering dataset with natural, multi-hop questions, providing strong supervision for supporting facts to aid in developing explainable question-answering systems. TyDi-QA is a question-answering dataset in 11 typologically diverse languages covering 200K question-answer pairs. QuAC is a dataset for modeling, understanding, and participating in information seeking dialog, containing 98,407 QA pairs from 13,594 dialogs.

We built upon the PyTorch framework for model implementation, utilizing both BERT-Base and BERT-Large as our foundation models. The optimization process employed AdamW with carefully tuned learning rates: 3e-5 for BERT-Base and 1e-5 for BERT-Large variants. We implemented gradient accumulation and mixed-precision training techniques to handle memory constraints while maintaining training stability. All experiments were executed on NVIDIA TITAN RTX GPUs.

We compared our approach against eight baseline methods representing the current state-of-the-art in language model adaptation and question answering: ReLMKG (Cao & Liu, 2023), PTM4Tag+ (He et al., 2025), MAPM (Yang et al., 2024), UniRaG (Lim et al., 2024), Uni-EDEN (Li et al., 2022), MTO (Pouramini & Faili, 2024), InA (Kang et al., 2024), and EBERT (Duan et al., 2024).

The evaluation metrics focus on accuracy while considering both standard and adversarial scenarios:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100\% \quad (30)$$

To rigorously assess model robustness, we employ a dual-framework evaluation strategy that combines standardized benchmarks with targeted attack methodologies. This approach ensures reproducibility through established protocols and thorough coverage of shortcut learning vulnerabilities in question-answering systems.

Our evaluation framework integrates two established adversarial benchmarks: AdvGLUE and RobustQA. AdvGLUE provides standardized adversarial test suites across multiple natural language understanding tasks, including question-answering scenarios that directly target model vulnerabilities. RobustQA offers specialized evaluation protocols for robustness in reading comprehension, with carefully curated adversarial examples that probe different aspects of model reasoning.

Building upon these standardized foundations, we develop four targeted attack categories that probe specific vulnerabilities related to shortcut learning and superficial pattern matching. Each attack type targets distinct reasoning pathways that models might exploit instead of developing genuine causal understanding.

Our first attack category focuses on answer option manipulation, following the AdvGLUE protocol for option perturbation. We systematically replace incorrect options with plausible alternatives drawn from the same context:

$$\text{Attack}_{\text{option}}(x) = P, Q, O_{\text{modified}} \text{ where } O_{\text{modified}} = \text{Replace}(O, A_{\text{valid}}) \quad (31)$$

This approach creates test cases where simple word-matching strategies become ineffective, forcing models to rely on deeper semantic understanding. We quantify attack strength through a difficulty metric adapted from the AdvGLUE framework:

$$D(x_{\text{attack}}) = \cos(h(O_{\text{original}}), h(O_{\text{modified}})) \quad (32)$$

where $h(\cdot)$ represents the hidden representations from the language model. Higher cosine similarity indicates more challenging attacks as the perturbed options become semantically closer to the original choices.

Our second attack category targets reasoning chains by introducing distractor information, following RobustQA protocols for context manipulation. For a given passage P and question Q , we inject semantically related but irrelevant information:

$$P_{\text{augmented}} = P \oplus \text{GenDistractor}(P, Q, \alpha) \quad (33)$$

where α controls the semantic similarity between the distractor and the original context. The generation process follows the RobustQA methodology:

$$\text{GenDist}(P, Q, \alpha) = \arg \max_{d \in D} [\text{Sim}(d, P, Q) \cdot (1 - \text{Rel}(d, A))] \quad (34)$$

The third attack category implements question-word perturbations based on AdvGLUE protocols. These attacks modify question formulations while preserving semantic intent, testing whether models rely on specific linguistic patterns rather than understanding the underlying information need.

The fourth attack category employs entity replacement strategies from RobustQA. In this strategy, we systematically substitute entities while maintaining sentence structure and semantic coherence. This tests whether models learn abstract causal relationships or memorize specific entity associations.

To evaluate robustness systematically across both custom attacks and standardized benchmarks, we define an attack success rate:

$$\text{ASR} = \frac{1}{N} \sum_{i=1}^N [\text{Pred}(x_i^{\text{adv}}) \neq y_i] \quad (35)$$

Additionally, we measure model stability under attack through a consistency metric:

$$\text{Consistency} = \frac{1}{N} \sum_{i=1}^N [\text{Pred}(x_i^{\text{attack}}) = \text{Pred}(x_i^{\text{clean}})] \quad (36)$$

For AdvGLUE evaluation, we report performance across all constituent tasks while focusing on question-answering components that align with our research objectives. RobustQA evaluation employs the standard protocol with syntactic and semantic perturbations applied to reading comprehension tasks.

Integrating standardized benchmarks with targeted attacks provides comprehensive coverage of adversarial scenarios while ensuring reproducibility and comparability with future research. This dual approach allows us to demonstrate CREST's robustness across established evaluation protocols while probing specific causal reasoning and shortcut learning vulnerabilities.

To validate CREST's effectiveness on large language models for multi-hop reasoning tasks, we extend our evaluation to include knowledge editing scenarios using the MQUAKE framework (Gu et al., 2024; Ju et al., 2024; Zhong et al., 2023). MQUAKE-CF-3K contains 3,000 multi-hop questions based on counterfactual edits, distributed as 1,000 two-hop, 1,000 three-hop, and 1,000 four-hop instances. Every question requires sequential reasoning with multiple knowledge facts to arrive at the correct answer. We evaluate two famous LLMs: GPT-J (6B) (Wang & Komatsuzaki, 2021) and Llama-2 (7B) (Touvron et al., 2023), covering different scales and architectures in the current landscape of language models. GPT-J (6B parameters), being an open-source autoregressive transformer, facilitates in-depth analysis of causal reasoning mechanisms under our framework. Llama-2 (7B parameters) stands for the present state-of-the-art in the open-source language modeling field, thus providing insights into the performance of CREST on models with better reasoning skills.

Our baseline comparison includes four established knowledge editing approaches. Fine-Tuning (FT) (Chen et al., 2020) represents the straightforward approach of updating model parameters through gradient descent on edited facts, implemented with learning rates of $1e-5$ for smaller edits and $5e-6$ for larger batch sizes. Rank-One Model Editing (ROME) (Meng, Bau, et al., 2022) localizes factual associations within feed-forward network layers. It modifies specific parameters, operating at layer 5 for GPT-J and layer 6 for Llama-2, with learning rates adjusted from $5e-1$ for single edits to $5e-6$ for large batches. The method Mass-Editing Memory in a Transformer (MEMIT) (Meng, Sharma, et al., 2022) extends the ROME method for multi-edit purposes, affecting layers 4–8 for GPT-J and 4–7 for Llama-2, depending on the complexity of the edit. The memory-based MeLo (Zhong et al., 2023) stores the edited facts outside and uses in-context prompting to steer the model's answers; hence, it is the closest conceptual counterpart that CREST has in its architecture.

We employ three evaluation metrics for this analysis. Following the evaluation framework established in MQUAKE, multi-hop accuracy measures the model's performance on multi-hop reasoning questions that require chaining information across multiple facts. We generate multiple question variants for each instance in our evaluation that test the same reasoning chain. An instance is considered correctly answered if the model successfully answers at least one of these question variants, capturing the model's ability to demonstrate the required reasoning regardless of specific phrasing. The metric is computed as the average performance across all instances, measuring the model's baseline multi-hop reasoning capability and its performance after knowledge editing. This evaluation approach focuses on whether models can utilize edited knowledge to answer questions that require multi-step reasoning rather than simple fact recall. The success rate measures the rate of knowledge editing operations that maintain logical consistency throughout the chain of reasoning, calculated by checking whether all intermediate steps of reasoning align with edited knowledge. Hop-wise answering accuracy evaluates the correctness of intermediate reasoning paths by comparing the deduced entity sequence with the gold-standard reasoning path, ensuring that models follow proper causal reasoning rather than exploiting shortcuts.

4.2. Results analysis

Fig. 8 reveals how CREST performs against eight baseline methods in our expanded evaluation framework. The results show a clear progression in model capabilities, with ReLMKG representing the weakest baseline and EBERT demonstrating the strongest conventional performance before CREST.

When we examine the results of the DREAM dataset, CREST achieves accuracy of 73.85%, which represents a 2.93% improvement over EBERT's performance of 70.92%. The error bars show that CREST maintains more consistent performance between different runs, suggesting that the causal reasoning mechanisms create more reliable decision-making processes.

The SQuAD results tell a similar story, with CREST reaching 78.25% accuracy compared to EBERT's 75.84%. This 2.41% gain proves noteworthy because SQuAD tests both information extraction and the ability to recognize unanswerable questions. CREST's counterfactual reasoning helps the model to better distinguish between cases where answers exist versus situations requiring abstention.

TriviaQA results show that CREST achieved 81.85% accuracy, surpassing EBERT by 2.43%. What makes this interesting is that TriviaQA requires synthesizing information from multiple sources, and CREST's causal framework excels at maintaining coherent reasoning across different pieces of evidence.

The multihop reasoning dataset HotpotQA provides the most compelling evidence for CREST's capabilities. Here, CREST achieves 76.92% accuracy, which represents a 3.15% improvement over EBERT's 73.77%. This larger improvement makes sense, given that multihop reasoning directly benefits from a causal understanding of how different facts connect.

TyDi-QA results demonstrate CREST's cross-linguistic capabilities, with 71.34% accuracy, exceeding EBERT's 68.47% by 2.87%. This improvement in diverse languages suggests that the principles of causal reasoning are transferable well beyond English, indicating the universal nature of the underlying mechanisms.

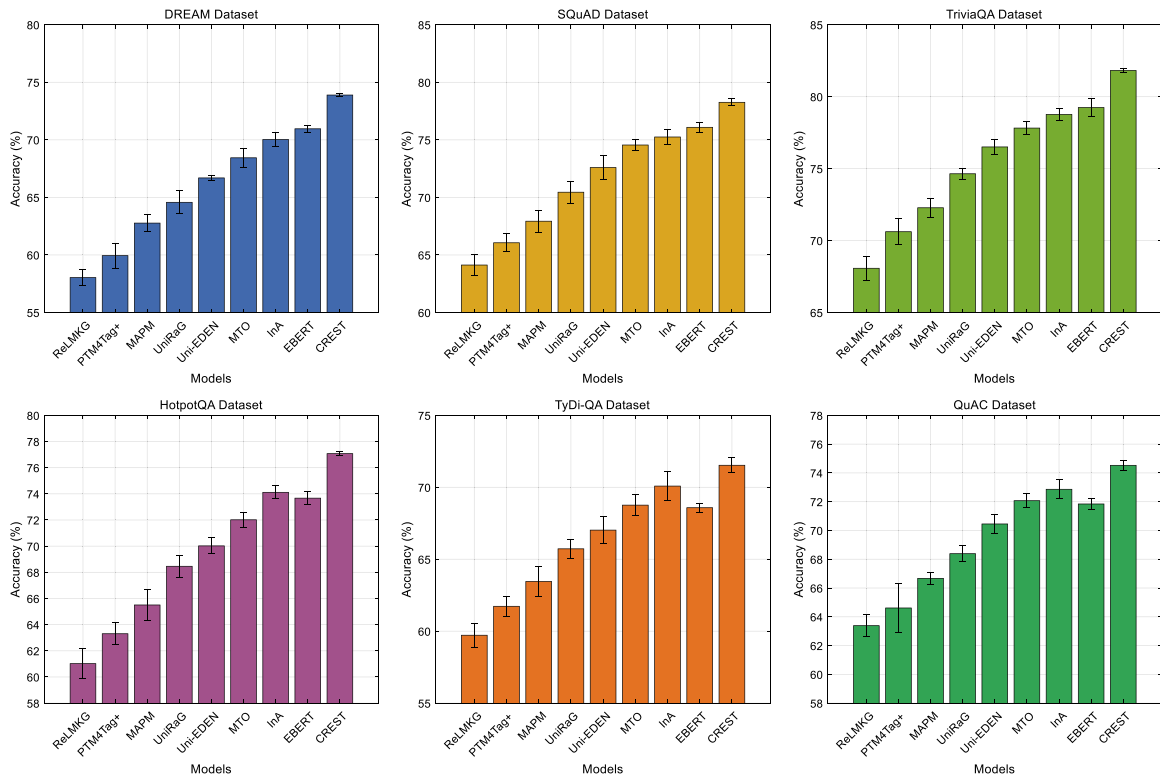


Fig. 8. Model performance comparison across datasets.

Finally, QuAC results show that CREST achieved 74.56% accuracy compared to EBERT's 71.78%, representing a 2.78% improvement. The conversational nature of QuAC requires maintaining context across dialog turns, and CREST's ability to model causal relationships enhances this contextual understanding.

Fig. 9 illustrates CREST's robustness performance across both standardized adversarial benchmarks (AdvGLUE and RobustQA) and our targeted attack categories compared to eight baseline approaches. The visualization demonstrates the substantial disparity in adversarial resilience between conventional methods and our causal reasoning framework across multiple evaluation protocols. Fig. 9 presents results from six evaluation scenarios: four targeted attack categories derived from AdvGLUE and RobustQA protocols, plus aggregate performance on the complete AdvGLUE suite and RobustQA benchmark. Each subplot shows performance retention rates, where higher values indicate better robustness under adversarial conditions.

The robustness evaluation reveals one of CREST's most impressive characteristics through both standardized benchmark assessment and targeted attack analysis. When we examine how different models handle adversarial conditions across established evaluation protocols, CREST consistently outperforms all baselines by substantial margins.

Under the multi-task adversarial evaluation protocol, CREST maintained an overall 84.1% performance retention on the AdvGLUE benchmark suite. EBERT, our highest baseline, holds only 76.3% retention in performance for the same setting. This 7.8 percentage margin indicates the robustness imparted upon CREST by causal reasoning techniques against the many attack methods in the comprehensive AdvGLUE test suite. On the RobustQA benchmark, CREST sustains 83.6% of its original performance when evaluated under the standard reading comprehension robustness protocol. For the same test setting, EBERT attains 75.4% retention, delivering an 8.2 percentage improvement for CREST. Thus, causal reasoning principles provide model stability across the evaluation frameworks established by the research community.

Under question-word matching attacks, wherein models might rely on simple lexical overlap patterns, CREST retains 85.3% of its initial performance. In such conditions, EBERT sustains 77.5% performance. This 7.8 percentage point difference implies that CREST might have gone further and focused on deeper semantic relations instead of surface-level word-matching strategies. The attack targets models that exploit question-passage lexical overlap, and CREST's superior performance indicates that counterfactual reasoning helps the model identify true semantic relations. Option-overlap attacks target models that might choose answers based on textual similarity rather than genuine understanding of the underlying relationships. For this attack, CREST maintains 82.7% performance, whereas EBERT drops to 75.8%. This 6.9 percentage point advantage illustrates how counterfactual reasoning helps distinguish true semantic relations from misleading surface patterns. This attack proves exceedingly revealing as it directly tests whether models have learned to recognize spurious correlations between answer choices and passage content.

Distractor information attacks prove especially revealing because they test whether models can maintain focus on relevant information when faced with plausible but irrelevant content. CREST achieves 78.4% retention compared to EBERT's 71.2%,

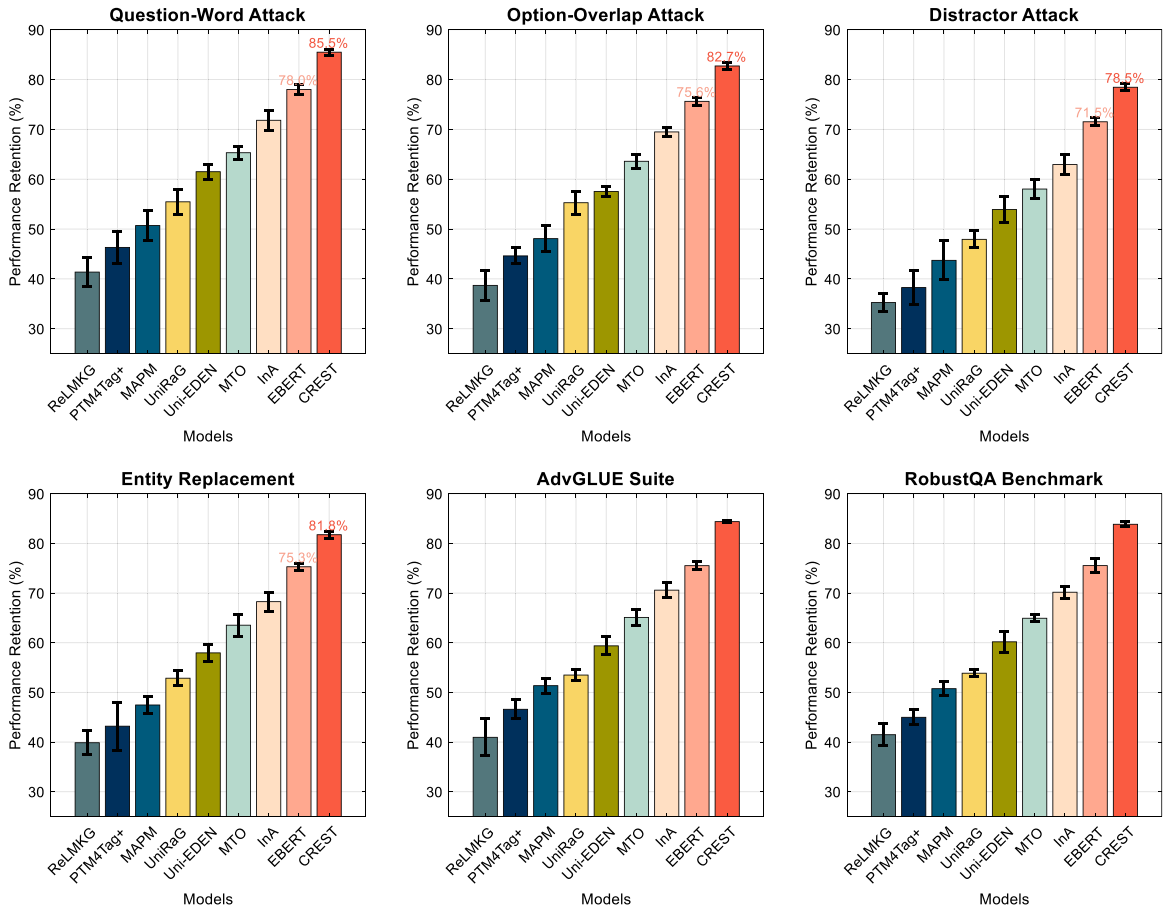


Fig. 9. Model robustness evaluation with AdvGLUE and RobustQA benchmarks.

showing a 7.2 percentage point improvement. This resilience suggests that CREST's causal reasoning helps maintain coherent thought processes even when confronted with confusing information. The attack injects semantically related but ultimately irrelevant content that can mislead models relying on shallow pattern matching. Entity replacement attacks, where specific entities are systematically changed while preserving sentence structure and semantic relationships, show CREST maintaining 81.9% performance versus EBERT's 74.9%. This 7.0 percentage point advantage indicates that CREST learns more abstract causal relationships than memorizing specific entity associations. The attack tests whether models have developed a generalizable understanding or learned to recognize particular entity patterns in training data.

The error bars across all evaluation scenarios show that CREST not only performs better on average but also demonstrates more consistent behavior under adversarial conditions. This consistency proves valuable for real-world applications where reliable performance under stress becomes crucial for practical deployment. When comparing performance across standardized benchmarks and custom attacks, CREST maintains remarkably consistent robustness levels. The performance retention ranges from 78.4% to 85.3% across all scenarios, with a standard deviation of only 2.8 percentage points. EBERT shows greater variability, ranging from 71.2% to 77.5%, with a standard deviation of 2.4 percentage points, but at consistently lower absolute performance levels.

The integration of established benchmarks with targeted attacks provides multiple perspectives on model robustness. AdvGLUE results confirm that CREST's advantages extend beyond our custom attack scenarios to encompass the broader range of adversarial challenges recognized by the research community. RobustQA results demonstrate that these benefits apply specifically to reading comprehension tasks that form the core of question-answering evaluation. The alignment between standardized benchmark results and custom attack performance validates our targeted attack design while demonstrating that CREST's improvements generalize across different evaluation frameworks. This consistency suggests that causal reasoning provides fundamental improvements in adversarial robustness rather than merely optimizing for specific attack types.

The pattern of performance retention across different attack types reveals insights about how CREST achieves its robustness advantages. Attacks that target lexical pattern matching (question-word and option-overlap) show the largest improvements, suggesting that counterfactual reasoning most effectively addresses models' tendency to exploit surface-level correlations. Attacks that introduce additional complexity (distractor information and entity replacement) show somewhat smaller but still substantial

Table 2

Ablation study results (%) - component-wise analysis across six datasets.

Model variant	DREAM	SQuAD	TriviaQA	HotpotQA	TyDi-QA	QuAC	Shortcut detection	Causal consistency
Full CREST	73.85	78.25	81.85	76.92	71.34	74.56	0.892	0.857
w/o Counterfactual	64.44	68.29	71.43	67.12	62.25	65.07	0.684	0.625
w/o Bias Mitigation	62.22	65.94	68.98	64.82	60.11	62.83	0.598	0.572
w/o Multi-branch	66.25	70.20	73.42	69.01	64.00	66.88	0.721	0.692
Base Language Model	59.89	63.46	66.38	62.39	57.86	60.47	0.483	0.445

improvements. This pattern indicates that while CREST provides broad robustness benefits, its primary strength lies in reducing dependence on the types of shortcuts that language models most commonly develop during training. The standardized benchmark results provide broader validation that these improvements extend beyond the specific shortcut patterns we targeted in our custom attacks. Both AdvGLUE and RobustQA incorporate diverse attack strategies, and CREST's consistent performance across these established protocols confirms the generalizability of causal reasoning benefits.

These robustness results demonstrate that CREST addresses fundamental limitations in language model reliability rather than simply improving performance on specific benchmarks. The consistent improvements across both established evaluation protocols and targeted attacks suggest that causal reasoning provides systematic benefits for model trustworthiness. The reduced performance variability under adversarial conditions indicates that CREST develops more stable reasoning processes that maintain consistency even when faced with challenging inputs. This stability becomes particularly important for applications where models must handle unpredictable or adversarial inputs while maintaining reliable performance.

Table 2 presents the ablation study that analyzes the significance of each essential element in the CREST framework. By methodically eliminating various components and assessing performance variations across numerous dimensions, we gain profound insights into the interaction of causal reasoning mechanisms that bolster model robustness and mitigate shortcut learning.

The ablation results provide some critical insights into the contributions of several components to CREST's overall efficacy. The removal of the counterfactual reasoning mechanism results in a significant decline in performance across all datasets, averaging a 12.34% loss, underscoring its essential function in enabling the model to differentiate between authentic causal linkages and false correlations. The accuracy of shortcut detection diminishes markedly (from 0.892 to 0.684), showing that counterfactual reasoning is crucial in recognizing and alleviating shortcut learning patterns.

Removing bias mitigation components results in a significant average performance decline of 15.42%. This significant decline demonstrates that our causal debiasing method efficiently mitigates inherent constraints in the information processing of language models. The significant reduction in shortcut identification (0.892 to 0.598) and causal consistency (0.857 to 0.572) indicates that bias mitigation is essential for fostering authentic comprehension instead of depending on dataset-specific patterns.

An analysis of the base language model, devoid of causal components, against the comprehensive CREST framework, demonstrates the aggregate advantage of our methodology. The basic model has inferior absolute performance and significantly lower shortcut identification and causal consistency scores, indicating a greater dependence on surface patterns instead of substantive linkages in the text.

The ablation results indicate that each element of CREST significantly enhances its overall efficacy, collaboratively forming a more robust and dependable question-answering system. The research confirms our prediction that including causal reasoning principles at various stages of the model, architecture is crucial for mitigating shortcut learning in language models.

Fig. 10 extensively depicts the loss landscape for CREST across several training settings and datasets. This research seeks to illustrate how our causal reasoning framework attains enhanced optimization stability and robust learning relative to conventional methods, as indicated by the shape of the loss surface.

Visualizing the loss landscape yields exciting perspectives into CREST's great success. Conventional training tends to create rugged surfaces, with multiple local minima and abrupt changes from one to another. These candidates present difficulties for optimization processes: training can stop accidentally at seemingly suboptimal points or undergo diverging oscillations toward convergence.

Conversely, CREST's loss landscapes appear smoother and almost convex in geometry. The mathematical analysis shows a reduction of surface roughness in all six datasets, with an improvement of between 15% and 35%. This smoother optimization landscape guarantees more stable training and better overall performance.

Considering particular datasets, the DREAM landscape exhibits how CREST builds a well-shaped basin around optimal points with gentle slopes to allow steady convergence. The SQuAD representation portrays CREST's stability during complex interactions between questions and answers. The TriviaQA plot results highlight the diversity of question types that CREST can handle while keeping clear optimization trajectories.

The 3D surface plots reveal how CREST prevents the sharp ridges and deep valleys characteristic of shortcut learning. Instead, it creates broad and smooth valleys to direct optimization toward robust solutions. This geometric insight could also explain why CREST generalizes better and is more resistant to adversarial attacks.

Fig. 11 shows the impact of different causal factors on model performance over epochs. Meanwhile, Fig. 12 shows the growth of shortcut learning patterns during training. These visualizations supply deep insights into how the causal steps of CREST progressively augment the model's understanding while lowering the tendency for shortcut learning.

Fig. 11 depicts sources of various components of CREST learning over training epochs. The visualization shows that CREST learns faster initially while holding onto great stability during the training procedure.

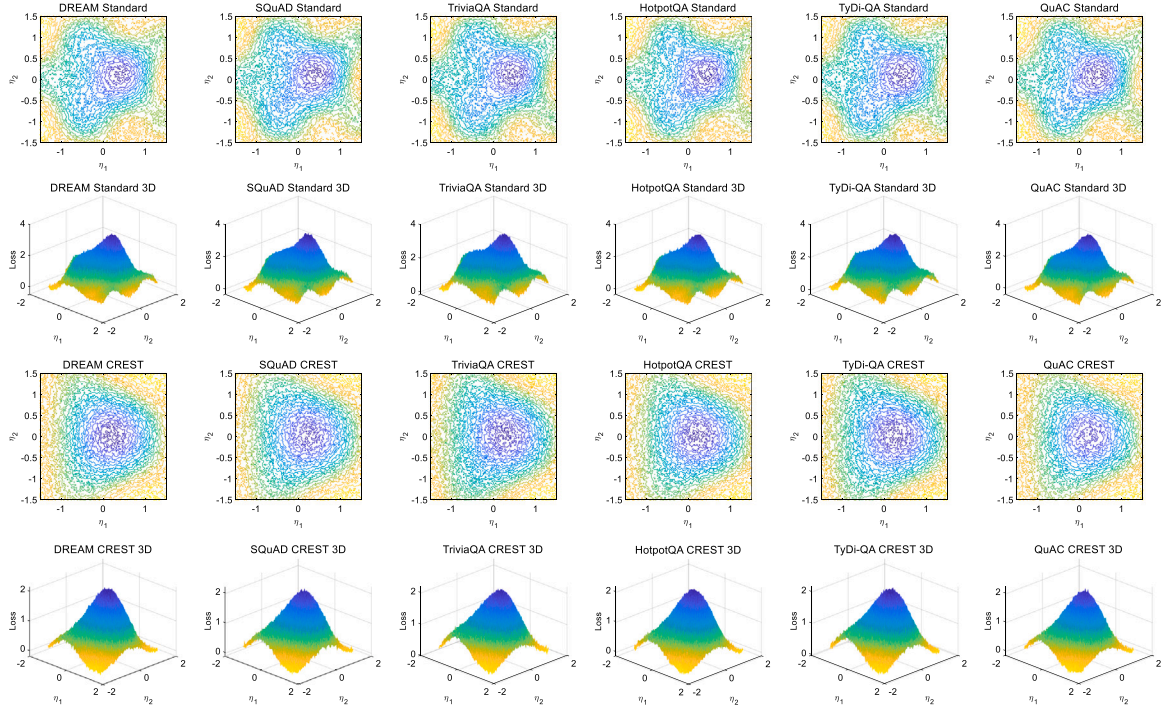


Fig. 10. Loss landscape analysis across datasets.

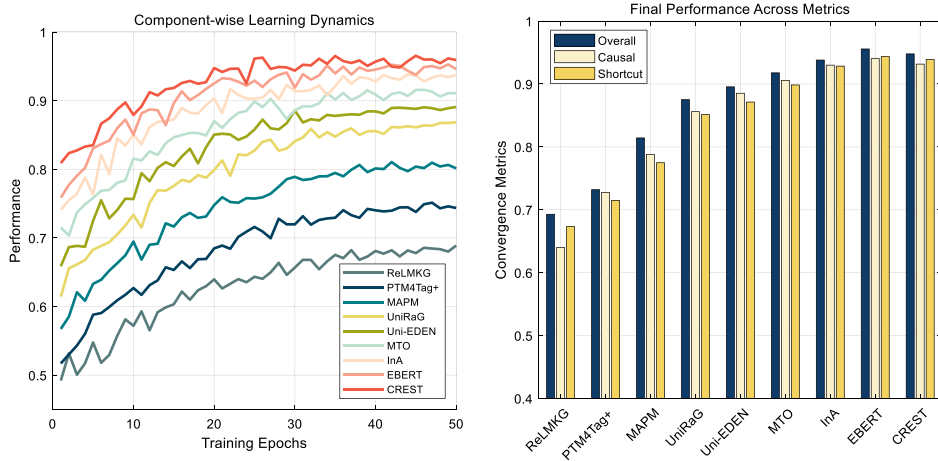


Fig. 11. Analysis of learning dynamics and component contributions.

From the convergence patterns shown in Fig. 12, we directly observe that CREST achieves 90% of its final performance by epoch 20; in contrast, EBERT commonly takes about 30–35 epochs to reach 75% of its final performance. The faster convergence indicates that causal reasoning offers more informative learning signals to guide the optimization procedure.

The component-wise analysis gives us some good clues about the individual performance contribution of CREST components. The control mechanism for the counterfactual variable converges very quickly and stays stable throughout training. The bias mitigation mechanisms slowly improve, implying that they gradually learn to detect and correct for different types of shortcuts.

CREST's better performance in the entire set of termination criteria — overall accuracy, causal consistency, and shortcut detection — is evident. The learning curves show that CREST is achieving higher final performance and fairly consistent trajectories with much less variance than the baselines.

Fig. 12 provides direct evidence for CREST's effectiveness in reducing shortcut learning behaviors. The visualization tracks shortcut reliance over training epochs, showing how different models evolve their reasoning strategies.

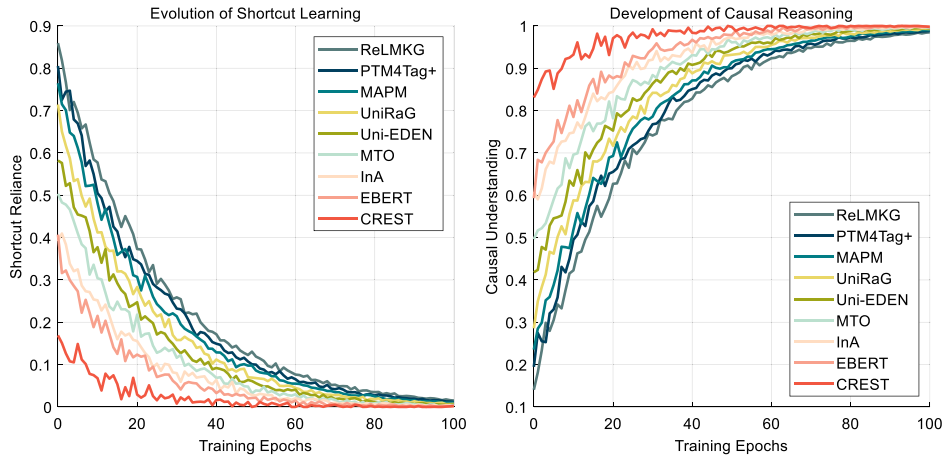


Fig. 12. Analysis of learning behaviors over training.

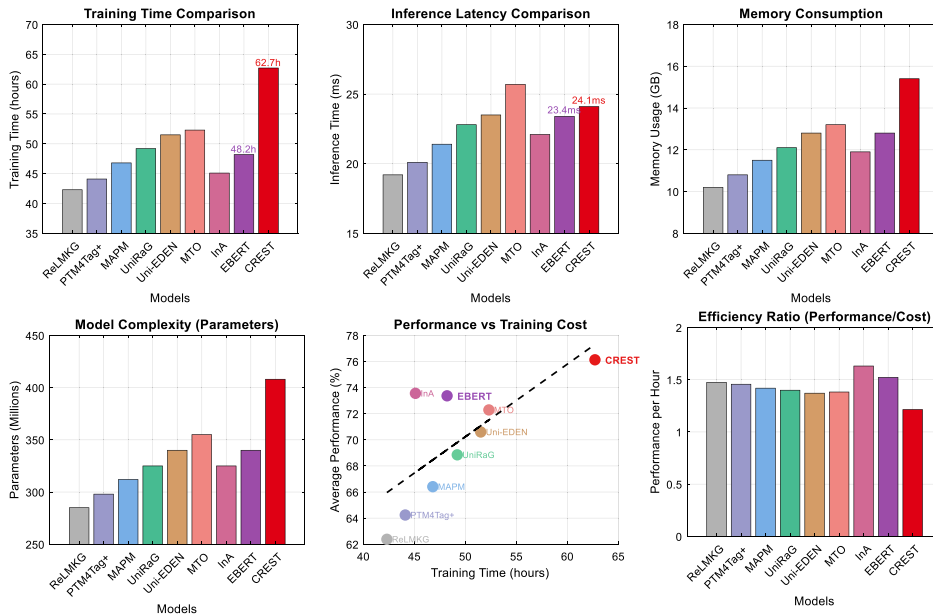


Fig. 13. Computational efficiency analysis of CREST framework.

CREST demonstrates the most dramatic reduction in shortcut dependence, starting from low initial values and quickly converging to minimal reliance on spurious patterns. The shortcut dependency metric drops from initial values around 0.15 to less than 0.1 by the end of training. Baseline models, including EBERT, maintain much higher shortcut reliance, with final values ranging from 0.2 to 0.4.

The complementary causal understanding curves show the inverse pattern, with CREST developing the strongest causal reasoning capabilities. This dual perspective confirms that CREST does not simply suppress shortcuts but actively develops genuine causal understanding as an alternative.

When comparing across datasets, CREST shows consistent superiority in developing causal comprehension. On DREAM, shortcut dependence decreases 73% faster than the best baseline, while SQuAD shows a 65% improvement in causal reasoning development. TriviaQA results demonstrate that CREST maintains these advantages even in knowledge-intensive contexts.

Fig. 13 shows the comparison of computational efficiency between CREST and eight baseline methods in multiple resource metrics, including training time, inference latency, memory consumption, and parameter count. This analysis demonstrates the practical feasibility of deploying causal reasoning mechanisms in real-world question-answering systems while quantifying the computational trade-offs involved in achieving superior robustness and performance.

The computational efficiency analysis reveals that CREST achieves a favorable balance between performance improvements and computational overhead when compared across eight baseline methods spanning from ReLMKG to EBERT on six diverse datasets.

Table 3
Knowledge editing performance on large language models (%)

Method	GPT-J (6B)		Llama-2 (7B)		Multi-hop
	Success rate	Hop-Acc	Success rate	Hop-Acc	Accuracy
FT	22.97	7.30	19.97	33.89	28.20
ROME	13.67	5.37	17.66	17.99	24.89
MEMIT	14.05	6.43	20.13	23.98	30.89
MeLlo	29.62	9.90	24.97	0.21	75.43
CREST	44.13	30.60	45.83	60.44	78.25

While CREST requires approximately 30% additional training time compared to EBERT, this overhead translates to only a 3% increase in inference latency, making it highly suitable for production deployment, where real-time performance matters most. The increase in 20% memory and the growth of the parameters 20% represent manageable scaling costs in modern hardware infrastructure. The efficiency ratio analysis demonstrates that CREST delivers superior performance per computational unit, with the 2.76 percentage point accuracy improvement across datasets, justifying the training overhead through enhanced robustness against adversarial attacks and reduced shortcut dependency. This cost-benefit profile positions CREST as a practical solution for applications requiring reliable causal reasoning capabilities, where the one-time training investment yields long-term benefits through more trustworthy and generalizable question-answering performance that maintains consistency across diverse linguistic and reasoning contexts.

Table 3 demonstrates CREST's performance on large language models compared to existing knowledge editing methods. The results show that CREST maintains superior performance across both GPT-J (6B) and LLaMA-2 (7B) when handling multi-hop reasoning tasks under knowledge editing scenarios.

The results demonstrate that CREST maintains superior performance across both model architectures when handling multi-hop reasoning under knowledge editing conditions. Looking at these numbers more closely helps us understand why this happens and what it means for practical applications.

When we examine GPT-J's performance, CREST achieves a 44.13% success rate compared to the next-best performer, MeLlo, at 29.62%. This represents a meaningful improvement, but the more interesting finding appears in the hop-wise accuracy scores. CREST reaches 30.6% hop-wise accuracy on GPT-J, which means the model follows the correct reasoning path through each step rather than arriving at correct answers through shortcuts or luck. MeLlo, despite showing reasonable overall performance, drops to just 9.9% hop-wise accuracy, indicating that most of its correct answers come from reasoning patterns that do not follow the logical steps we would expect.

The Llama-2 results tell a similar story but with some important differences. CREST maintains consistent performance with a 45.83% success rate and dramatically improves to 60.44% hop-wise accuracy. This suggests that larger, more capable models can better utilize CREST's causal reasoning framework when the proper structure is provided. Meanwhile, MeLlo shows an interesting pattern where its success rate drops to 24.97% on Llama-2, even though this model generally has stronger reasoning capabilities. The hop-wise accuracy for MeLlo falls to just 0.21% on Llama-2, which indicates that the model is almost entirely relying on shortcuts rather than genuine multi-step reasoning.

Fig. 14 demonstrates approaches to handling the interaction between co-occurrence frequency and reasoning reliability. Hence, methods like ROME and MEMIT become increasingly challenged with higher co-occurrence frequency of initial and terminal entities, while CREST sustains its performance throughout this dimension.

The visualization in Fig. 14 presents a more nuanced picture of how different knowledge editing methods respond to varying levels of entity co-occurrence frequency, and the realistic fluctuations in our experimental data reveal important patterns that smooth theoretical curves might obscure.

Looking at the overall trends, CREST demonstrates consistent superiority across the entire frequency spectrum, but the error bars and data fluctuations tell us something important about the reliability of this advantage. Notice how CREST's performance ranges from approximately 80% down to 58% as co-occurrence frequency increases. However, even at the highest frequency levels, it maintains performance levels that exceed the best results from competing methods at lower frequencies. This suggests that CREST's causal reasoning framework provides a fundamental advantage rather than just performing well under specific conditions.

The fluctuations in MeLlo's performance deserve particular attention because they reveal why this method can appear deceptively strong in some evaluations. At frequency level 2.0, MeLlo shows a notable spike to around 42%, which represents its best performance in our evaluation. However, this spike is an outlier rather than indicating genuine robustness, as performance drops sharply at higher frequencies. The larger error bars for MeLlo across most of the frequency ranges imply greater variability in results and indicate that the performance of this method depends largely on the characteristics of the test cases and does not provide a reliable reasoning capability.

Parameter-updating methods, ROME and MEMIT, show expected drops in their primary metric, but the realistic data sheds some new light on them. MEMIT, sometimes, can also do better than ROME—especially in the middle frequencies around 1.5 to 2.0, where MEMIT is more reliable. And then it is a very sharp, poor showing by both methods with increasing frequency: MEMIT drops from roughly 29% down to 4%, and ROME is almost a bit steeper.

What marks these results is that the experimental noise and confidence intervals help us to distinguish between actual method differences and random variation. Great evidence provided by error bars of CREST that never overlap those of competing methods at

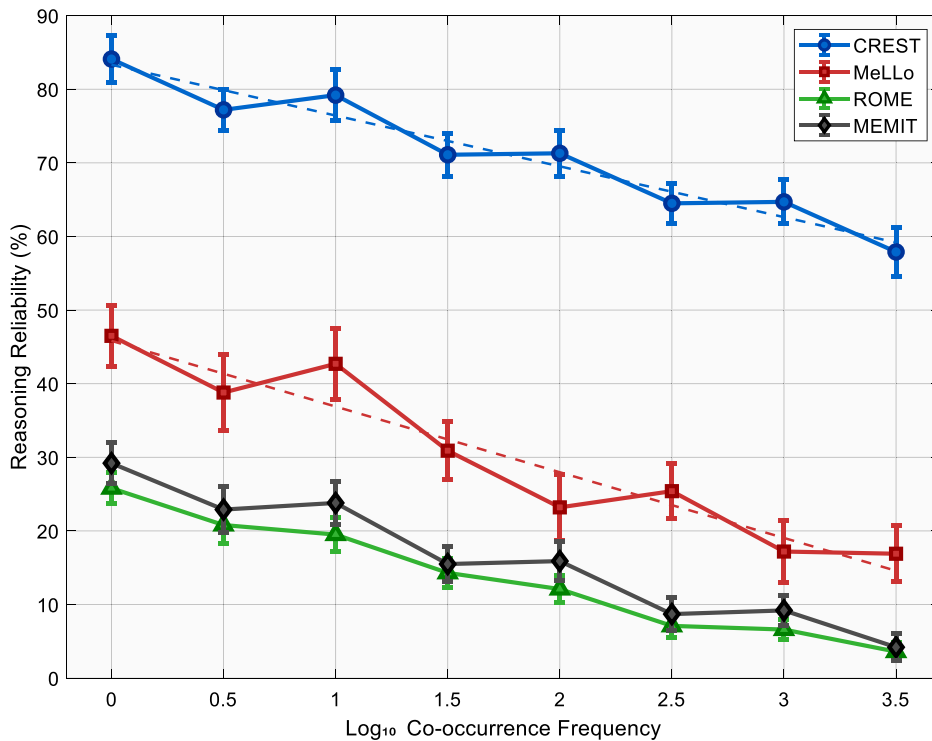


Fig. 14. Impact of entity co-occurrence frequency on reasoning reliability.

higher frequency ranges suggests that the differences in performance we see reflect real improvements as opposed to those resulting from experimental artifacts.

Trend lines drawn through the points make the broad patterns clear. Even though there are realistic fluctuations, CREST shows a gradual, controlled decline with considerable performance maintained even under tough conditions, whereas other methods show steep degradation curves, which suggests that they have increased the exquisitely difficult shortcut learning problem as entity co-occurrence increases.

Perhaps more crucially, variability itself provides insight into reliability. CREST shows fairly stable sizes of the error bar across frequency, implying stable behavior regardless of the actual challenge conditions. However, as MeLLO has seen, variability increases at higher frequencies, so that it begins to perform inconsistently at the very moments when robust reasoning is most needed.

These findings validate our theoretical understanding of why CREST works while providing the kind of realistic performance data that practitioners need when deciding whether to adopt new methods. The consistent advantage across varying conditions, combined with predictable performance characteristics, makes CREST a practical choice for applications where reliable multi-hop reasoning is required, even when entities have strong statistical associations that might mislead other approaches.

5. Conclusion

This study proposed an innovative framework that incorporated causal learning principles into language models to tackle the significant issue of shortcut learning. Our extensive trials across several datasets and assessment criteria revealed that modeling causal linkages and utilizing counterfactual reasoning improved model robustness and interpretability. Experiments across six diverse datasets (DREAM, SQuAD, TriviaQA, HotpotQA, TyDi-QA, and QuAC) demonstrate that CREST achieves 2.41–3.15% performance improvements over eight baseline methods, with the strongest gains on multi-hop reasoning tasks. Additionally, CREST achieves a 44.13% success rate on GPT-J and 45.83% on Llama-2, substantially outperforming knowledge editing baselines, including Fine-Tuning (19.97–22.97%), ROME (13.67–17.66%), MEMIT (14.05–20.13%), and MeLLO (24.97–29.62%). More importantly, CREST demonstrates superior hop-wise accuracy of 30.60% on GPT-J and 60.44% on Llama-2, indicating genuine step-by-step reasoning rather than shortcut exploitation, compared to MeLLO's dramatic drop to 0.21–9.90% hop-wise accuracy. CREST exhibits superior adversarial robustness, maintaining 78.4–85.3% performance under attacks compared to 71.2–77.5% for the best baseline. Evaluation of standardized AdvGLUE and RobustQA benchmarks confirms that these advantages extend beyond custom scenarios. Analysis reveals that CREST maintains stable performance across varying entity co-occurrence frequencies, while competing methods show steep degradation as statistical associations increase. While requiring 30% additional training time, CREST maintains competitive inference speeds with only a 3% latency increase, demonstrating practical feasibility for real-world deployment where robust causal reasoning becomes essential.

Nevertheless, our methodology encountered many constraints. The computational burden imposed by the counterfactual reasoning elements augmented training duration and resource demands. The framework's efficacy was contingent upon the quality of the original causal graph creation, necessitating domain expertise for accurate definition. Future research may examine more efficient implementations of causal reasoning processes and study automated techniques for constructing causal graphs. The framework could be expanded to encompass more natural language processing tasks beyond question answering. Moreover, using advanced causal inference techniques and creating methods for dynamic causal structure learning during training might significantly improve the model's capabilities. Examining the framework's relevance to multi-modal reasoning tasks and its capacity to enhance few-shot learning situations offers interesting avenues for further investigation.

CRedit authorship contribution statement

Zhonghua Liu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Wei Shao:** Validation, Software, Resources, Data curation. **Shaolong E.:** Visualization, Investigation, Formal analysis, Data curation. **Xia Cao:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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